An Interactive Mixed Reality Imaging System for Minimally Invasive Surgeries

Samir A. El-Seoud
Faculty of Informatics and Computer Science (ICS)
The British University in Egypt (BUE)
El Sherouk City, Misr-IsmaliaRoad, Cairo-Egypt
samir.elseoud@bue.edu.eg

Amr S. Mady
Faculty of Informatics and Computer Science (ICS)
The British University in Egypt (BUE)
El Sherouk City, Misr-IsmaliaRoad, Cairo-Egypt
amr122437@bue.edu.eg

Essam A. Rashed
Faculty of Informatics and Computer Science (ICS)
The British University in Egypt (BUE)
El Sherouk City, Misr-IsmaliaRoad, Cairo-Egypt
essam.rashed@bue.edu.eg

ABSTRACT

In orthopedic surgery, it is important for physicians to completely understand the three-dimensional (3D) anatomical structures for several procedures. With the current revolution in technology in every aspect of our life, mixed reality in the medical field is going to be very useful. However, medicine has a visualization problem hindering how surgeons operate. The surgeons are required to imagine the actual 3D structure of the patient by looking at multiple 2D slices of the patients' body. This process is time consuming, exhausting and requires special skill and experience. Moreover, patients and surgeons are exposed to extra x-ray doses.

Therefore, it is important to provide the surgeon with a better way to diagnose the patient; a way that is more accurate and locates where the problem is in a faster and more efficient manner. Medical imaging systems usually provide 3D images that can guide interventional clinical procedures. However, it is difficult to map the 3D anatomical structure with real objects. This project investigates and solves this problem by providing a mixed reality technology solution that merges the 3D image with real objects to facilitate the work progress of the surgeon. The proposed solution is an interactive mixed reality (MR) system for minimally invasive surgeries. The system is based on mapping the patient volume scan using computed tomography (CT) or Magnetic Resonance Imaging (MRI) to a 3D model of the patient's body. The rendered model can be used in MR system to view 3D human structures through a set of wearable glasses.

CCS Concepts

• **Computing methodologies→Image processing.**

Keywords

component: mixed reality, volume rendering, medical imaging,

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1. INTRODUCTION

The current evolution in medicine and technology should proceed at the same level. Furthermore, medicine should take advantage in the speedily development in technology. One of these significantly important parts of tech-medicine applications are the visualization of human anatomy [1]. Interventional radiology procedures using imaging guidance such as CT/MRI does not meet 100% surgeon's satisfaction. In current procedures, radiologists must scan the patient from different positions. Thereafter, surgeons and radiologists must investigate the scanned images to better locate the problem. Consequently, doctors and patients are exposed to heavy radiation. However, some types of medical imaging systems provide a series of scans that can be viewed as a 3D model using appropriate software that can guide interventional clinical procedures.

Our developed system should be a one step forward to solve the problem of visualizing human bodies.

Volume rendering of three-dimensional (3D) image data of patient's multiple slices is the revolution in imaging human body [2]. A voxel is the 3D equivalent to a pixel and are the smallest element in a 3D object [3]. Voxels are used to build 3D objects, mostly used in computer graphical applications like computer games, but also used to render a volume. Applications on volume rendering have taken a large part in interventional and minimally invasive surgeries over the past couple of years. Before volume rendering there were other techniques that concentrate on visualization via surface shading:

- It transforms the volumetric data into geometric primitives then screen the pixels
- It is good but not the best for visualization.

When it comes to volume rendering, the technique displays the information inside the volume, it is a direct display, the technique transforms volumetric data to screen pixels directly, and also it uses transparency to see through volumes.

In 2008, researchers from university of München, Germany, started a project that maps the CT scans obtained with patient body. Their Augmented Reality (AR) system of optical tracking and video see-through head mounted device (HMD) for visualization was developed to keep track of the objects in the scene. This process is carried out by two separate optical ray-casting.

tracking systems. Four infrared ARTtrack2 cameras have been mounted to the room's ceiling to obtain an outside-in optical tracking system, while an IR camera mounted directly on the HMD is used as an inside-out optical tracking system [4].

This work aims to develop a MR software that will be used in minimally invasive surgeries and interval procedures. Our goal is to reduce the heavy load of scans visualization, as well as saving time and effort. This procedure will be much cheaper than the previous methods. Our system requires only a smartphone and a MR ready headset. Mixed Reality is the combination of Virtual Reality (VR) and AR [5]. This combination brings together the real world and the digital one in one reality [6, 7]. It allows the users to interact with both physical and virtual items.

2. MATERIAL AND METHODS

2.1 System Overview

The introduced system visualizes medical images (CT, MRI) as a 3D object. First, we use the developed software to visualize the medical images by using volume rendering ray-casting technique. The term volume rendering is used to describe techniques which allow the visualization of three-dimensional data. Volume rendering is a technique for visualizing sampled functions of three spatial dimensions by computing 2D projections of a colored semitransparent volume. The technique works as follows:

Step 1:

- Trace from each pixel a ray into object space
- Compute and accumulate color/opacity value along the ray
- Assign the obtained value to the pixel

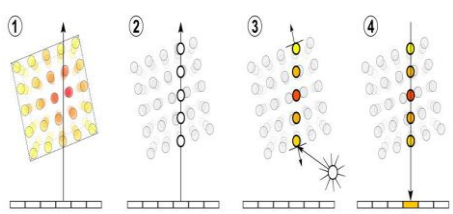

Fig.1 How ray-marching works [8, 10].

Step 2:

In this step, we use compositing (alpha blending), i.e. the iterative computation of discretized volume integral. Figure 2 illustrates on how alpha blending works while each ray goes through the object on its direction.

The developed software runs on Samsung Gear VR headset, and using its pass-through camera feature. It will enable the software to augment the 3D object of the medical scans in real world space. Interactiveness with the augmented object will be performed using the Gear VR controller. Users could manipulate the viewed 3D object generated from the medical images sliced by hiding parts of the object or view it in different ways with some GUI features to help the user to interact with it more easily, such as:

- Increasing visibility
- Increasing and Decreasing Opacity
- Clipping (removing parts of the object) on the X, Y and Z axes.
- Rotation and Translation

Breifly, the considered scenario may be summarized as follows:

1. First obtain volumetric medical data (DICOM/RAW).
2. Preprocess the data to the best possible losslessly form of useable data.
3. The data are stored afterwards on a smartphone then mount it on a VR headset that has a passthrough camera feature.
4. The software will render the preprocessed data as a 3D object into reality using augmented reality technology through the virtual reality headset.
5. User will interact with the 3D object via Gear VR controller.

Concluding this scenario, surgeons and radiologists will be able to see the scanned slices of the patient as a real 3D object in front of them and will be able to interact with it through a controller. They have the capability to zoom in or out or even to make parts of the object transparent as well as clipping parts of it.

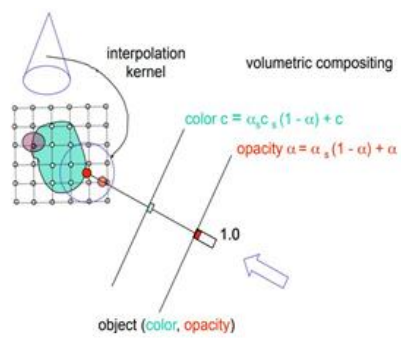

Fig. 2 Compositing of pixels' color/opacity along the ray [9, 10].

2.2 Image Acquisition

CT or MRI scanners first scan the patient. Afterwards, data is sent to an online archive to be stored and registered. Thereafter, the data has to be sent to the smartphone via wireless for processing.

3. RESULTS AND DISCUSSION

3.1 Results

The figures below are samples from the first dataset (221 slices).

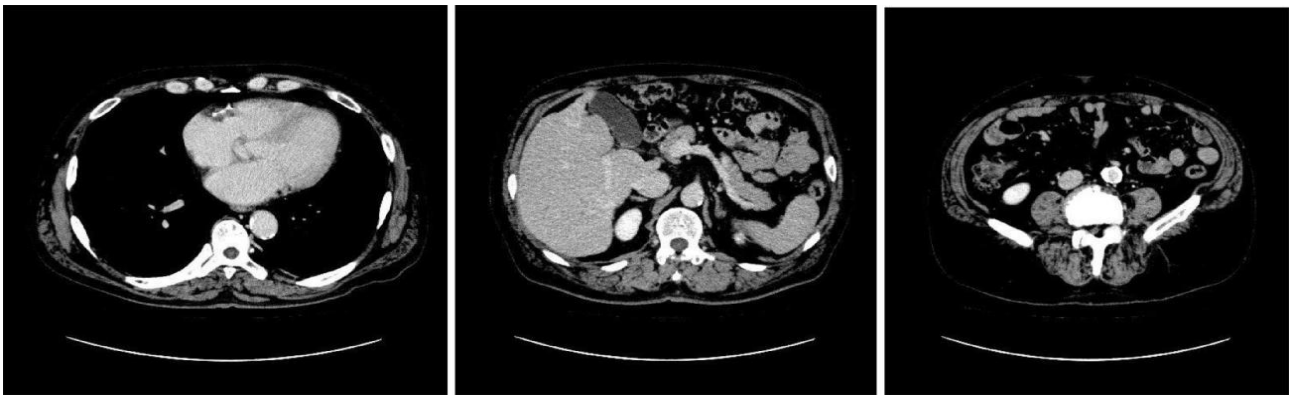

Fig3.a, b, c samples from the first used CT scans dataset

In Fig. 4(a-c), the images show the surface of the patient's body as a 3D object viewed from three different angles using the first dataset.

Fig. 4a (left) A 3D object rendered with ray-marching by using the first dataset, full opacity, no clipping, front facing, Fig. 4b (center) rotated 90 degrees on the Y-axis, Fig. 4c (right) rotated 90 degrees on X-axis

In Fig. 5 (a-c), showing the inside of the body after decreasing the opacity viewed from three different angles using the first dataset.

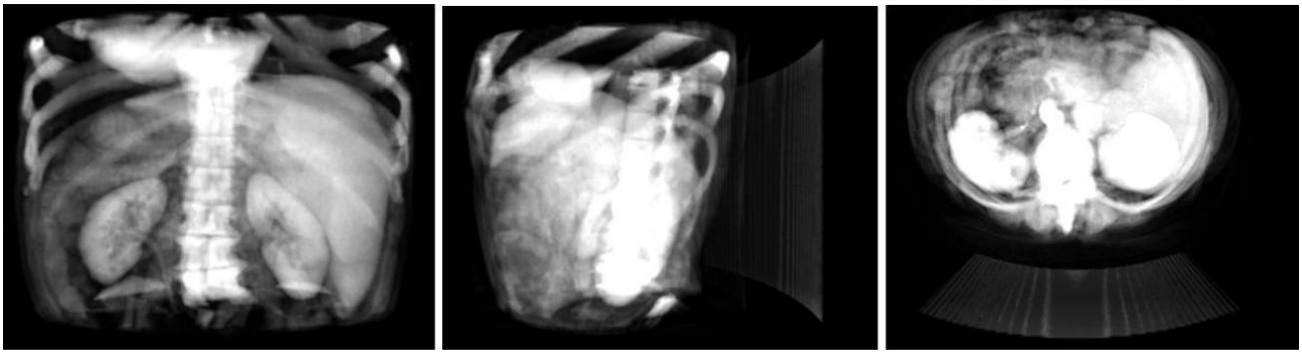

Fig. 5a (left) Same 3D object, 0.03 opacity, no clipping, front facing, Fig. 5b (center) rotated 90 degrees on the Y-axis, Fig. 5c (right) rotated 90 degrees on X-axis

In Fig. 6 (a-c), showing only half of the rendered object viewed from three different angles using the first dataset.

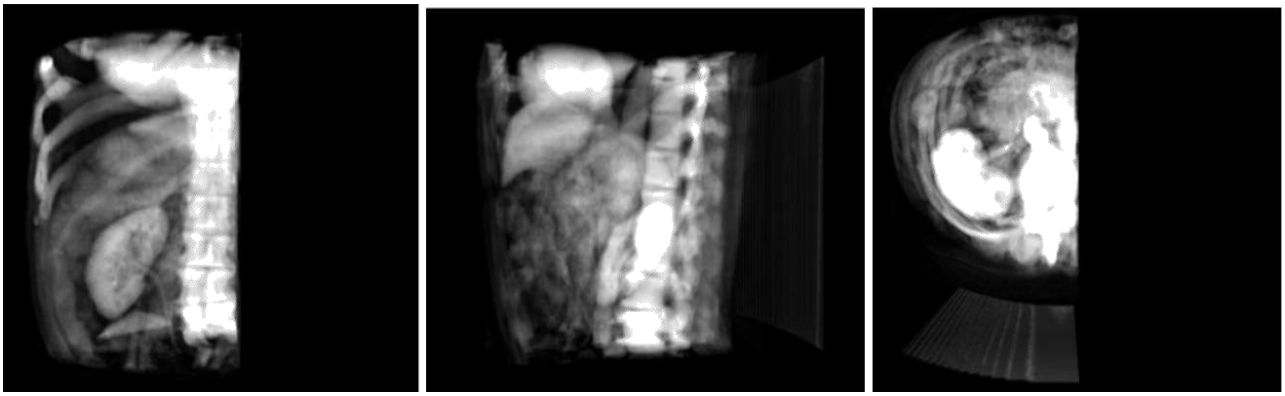

Fig. 6a (left) Same 3D object, 0.03 opacity, clipped 50% of it on X-axis, front facing, Fig. 6b (center) rotated 90 degrees on the Y-axis, Fig. 6c (right) rotated 90 degrees on X-axis.

The figures below are samples from the second dataset (361 slices).

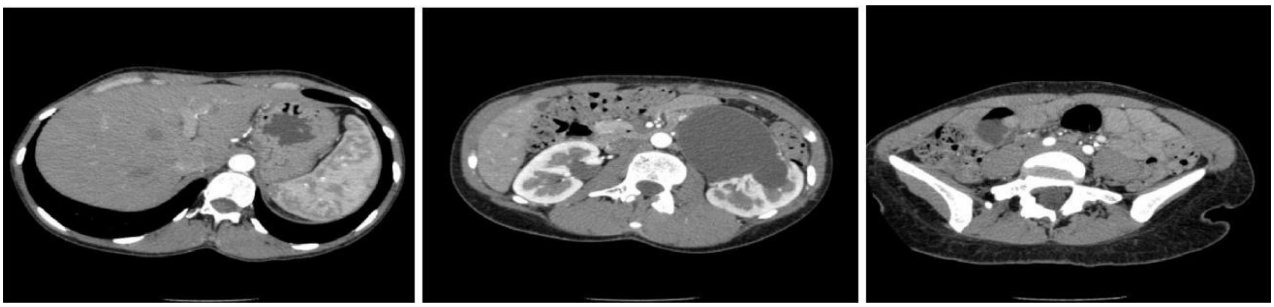

Fig7.a, b, c samples from the second used CT scans dataset

In Fig. 8(a-c), the images show the surface of the patient's body as a 3D object viewed from three different angles using the second dataset.

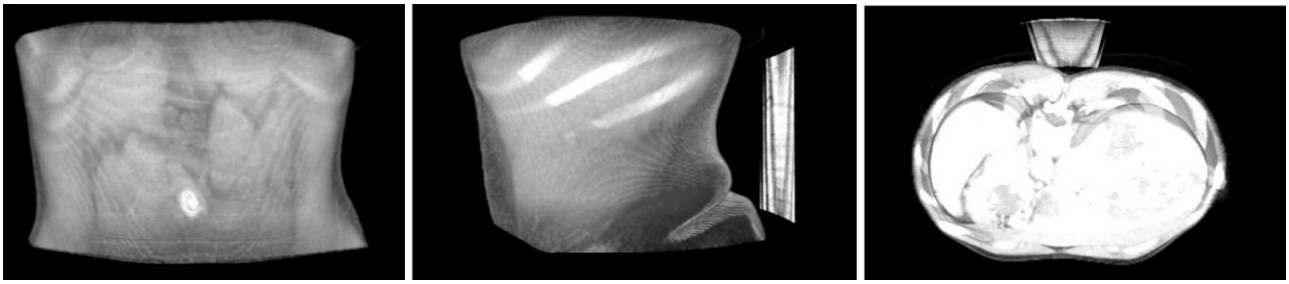

Fig. 8a (left) A 3D object rendered with raymarching using the second dataset, full opacity, no clipping, front facing, Fig. 8b (center) rotated 90 degrees on the Y-axis, Fig. 8c (right) rotated 90 degrees on X-axis.

In Fig. 9 (a-c), showing the inside of the body after decreasing the opacity viewed from three different angles using the second dataset.

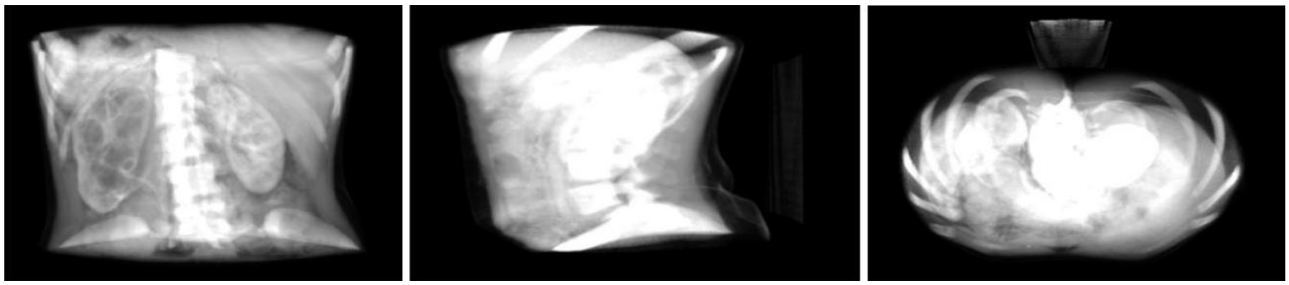

Fig. 9a (left) Same 3D object, 0.03 opacity, no clipping, front facing, Fig. 9b (center) rotated 90 degrees on the Y-axis, Fig. 9c (right) rotated 90 degrees on X-axis.

In Fig. 10 (a-c), showing only half of the rendered object viewed from three different angles using the second dataset.

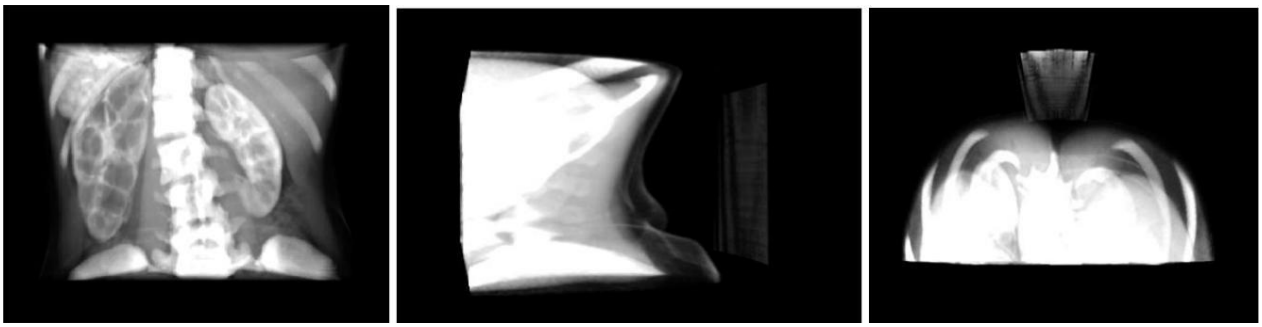

Fig. 10a (left) Same 3D object, 0.05 opacity, clipped 50% of it on Y-axis, front facing, Fig. 10b (center) rotated 90 degrees on the Y-axis, Fig. 10c (right) rotated 90 degrees on X-axis.

3.2 Discussion

The proposed software will help in minimizing the visualization of medical images, saving time and effort for surgeons and radiologist, with relevantly fast run time. It requires few minutes to render a data-set of 300 images.

4. CONCLUSION.

In this paper, we discussed the developed software that will be used as a new method in visualization of medical images. This software can deliver better visualizations to surgeons and radiologists, helping to create a better environment for surgeries.

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